Republic of the Philippines
Department of Environment and Natural Resources
MINES AND GEOSCIENCES BUREAU
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LANDSLIDE AND FLOOD THREAT ADVISORIES

To: Brgy. Chairperson
Brgy. Malanday
City of Marikina
Metropolitan Manila

Date: Oct 9, 2024

Dear Sir/Madam:

Please be advised that the Geohazards Mapping and Assessment Team (GMAT) of the Mines and Geosciences Bureau (MGB) has conducted **landslide and flood assessment** for the updating of 1:10,000 Scale Geohazard Map in your barangay. The following are the results and recommendations following the assessment:

Purok / Sitio / Area of Concern	GPS Coordinates (Coordinate System: Luzon 1911)	Flood Susceptibility Rating	Geohazards Assessment
Libis Bulelak	14°39'14.8" N 121°05' 45.5" E	**Very High**	Brgy. Malanday is located on the eastern side of the Marikina river, within an inner bend. This makes the area more susceptible to flooding. Most of the area was inundated during Typhoon Ondoy and Typhoon Ulysses. In Sitaw-Libis-Buldak the area, the riverwall height was increased to help mitigate the flooding. Some areas experience as high as 7 m flood height (half of 3rd floor) and it took 1.5 days before subsiding.
Libis Minahan	14°39' 15.1" N 121°05' 37.6" E	**Very High**	
Parkland Basketball	14°38' 52.1" N 121°05' 36.3" E	**Very High**	
Santa Teresita	14°38' 52.4" N 121°05' 52.7" E	**High**	

Remarks / Recommendations:

- ✓ **Implement pre-emptive evacuation** by the City Disaster Risk Reduction Management Office (CDRRMO) and Barangay Disaster Risk Reduction Management Office (BDRRMO) during monsoon (heavy rainfall) and typhoon seasons.
- ✓ **Construction and/or improvement of drainage system**. Existing channels should be broadened and deepened to ensure smooth drainage of floodwaters/runoff. Avoid improper waste disposal on channels as it may cause clogging and obstruction of water flow. Design of canals should also consider obstructions by buildings/infrastructures.

- ✓ **Strengthen and improve** flood warning system with corresponding warning levels for pre-emptive evacuation purposes
- ✓ Adhere to the warnings given by proper authorities prior to the landfall or passing of typhoon and other prevailing weather disturbances. This may include CDRRMO, Office of the Civil Defense, and other related agencies.
- ✓ **Construction of appropriate and well-studied engineering measures** (e.g. dike or flood wall etc.) should be implemented to prevent overtopping of floodwaters and the fast rate of erosion along rivers/creeks. Make sure that the structure follows engineering standards and has structural integrity. However, even though there are engineering measures, communities near the river/creeks must remain vigilant and not complacent during strong typhoons and inclement weather.
- ✓ The barangay should have **constant communication with nearby/upper barangays** during typhoon and monsoon season. This is to gather information from their counterparts in terms of the flood situation in their areas. Usually, floodwaters coming from nearby/upper barangays are the main cause of flooding within the barangay area. This may serve as preparation for evacuation of their constituents, especially those near rivers/creeks/canals.
- ✓ The CDRRMO, in coordination with the BDRRMO, should conduct information and education campaigns on flood hazards to increase public awareness, especially to identified flood-prone communities.

Prepared by:

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Received by:

Designation:

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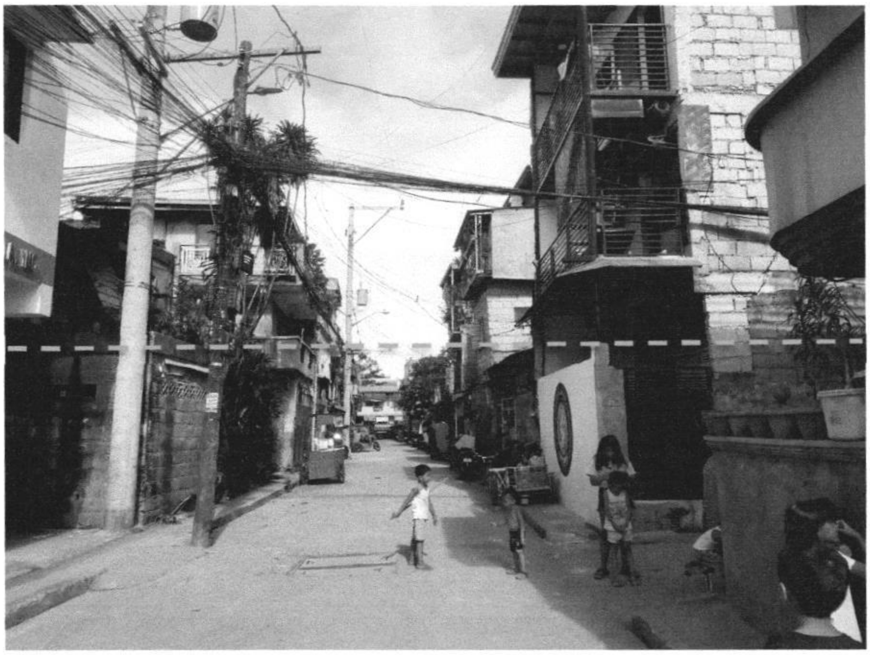
PHOTOS

Floodwall height increased in some areas of the barangay to help mitigate flooding.



Flood height during typhoon Ulysses. During Typhoon Ondoy the electrical posts were complete submerged.

Recommendations:

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- ✓ **Construction and/or improvement of drainage system**. Existing channels should be broadened and deepened to ensure smooth drainage of floodwaters/runoff. Avoid improper waste disposal on channels as it may cause clogging and obstruction of water flow. Design of canals should also consider obstructions by buildings/infrastructures.
- ✓ **Strengthen and improve** flood warning system with corresponding warning levels for pre-emptive evacuation purposes
- ✓ Adhere to the warnings given by proper authorities prior to the landfall or passing of typhoon and other prevailing weather disturbances. This may include CDRRMO, Office of the Civil Defense, and other related agencies.
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- ✓ The CDRRMO, in coordination with the BDRRMO, should conduct information and education campaigns on flood hazards to increase public awareness, especially to identified flood-prone communities.